

# Andrea Capone

PhD Student | Developer

✉ capone.andrea195@gmail.com | [in](#) andrea-capone | [G](#) andrew | [Globe](#) andrea-capone-personalwebsite

## TECHNICAL SKILLS

- Full-stack Developer
- Machine Learning
- Cloud Computing
- Multi-agent systems
- Data Analysis
- Database

## PUBLICATIONS

- *Reasoning About Intuitionistic Alternating-Time Temporal Logic*. In Proceedings of the **8th International Workshop on Artificial Intelligence and fOrmal VERification, Logic, Automata, and sYnthesis (OVERLAY)**, 2026.
- *GameL: a Neuro-Symbolic Agentic Architecture Toward Autonomous and Resilient Energy Communities*. In Proceedings of the **41st Italian Conference on Computational Logic (CILC)**, 2026.
- *A Neurosymbolic Approach for Landslide Risk Assessment*. In Proceedings of the **6th National Conference on Artificial Intelligence (Ital-IA)**, 2026.
- *Gaussian Process and Kalman Baselines for Bradyseismic Signal Extraction from GNSS Time Series at Campi Flegrei*. In Proceedings of the **XXXIV Italian Workshop on Neural Networks (WIRN)**, 2026.
- *Machine Learning and Explainable Deep Learning for Forecasting Landslide Deformation using EGMS Time Series*. Preprint of the Journal **Applied Soft Computing**, 2026
- *An Intuitionistic Version of Computation Tree Logic*. In Proceedings of the **22nd European Conference on Multi-Agent Systems (EUMAS)**, 2025.
- *An Intuitionistic Version of Alternating Time-Temporal Logic*. In Proceedings of the **22nd International Conference on Principles of Knowledge Representation and Reasoning (KR)**, 2025.

## PROJECTS

### PhD Project: Neurosymbolic AI for Bradyseism Risk Assessment

Nov 2024 – Present

National PhD in AI for Agrifood and Environment

Python, Machine Learning, Formal Methods

- Analysis of **GNSS time series** to model ground deformation trends in the Campi Flegrei area
- Study of **sinkholes and landslides** to identify additional high-risk areas across the region
- Application of **neurosymbolic approaches** integrating **Explainable AI** with formal logics to validate seismic risk scenarios
- Collaboration with **INGV** to provide decision-support tools for risk mitigation and resilience strategies

### Medical Survey System

Feb 2025 – Present

Interactive Surveys for Medical Conferences

JavaScript, Firebase

- Development of an **interactive survey platform** tailored for medical conferences, enabling real-time data collection and visualization
- Integration with **Firebase** for secure authentication, database management, and real-time synchronization
- Implementation of responsive front-end components in **JavaScript**, ensuring usability across multiple devices

### VITAMIN Model Checker

Feb 2024 – Present

Logic for Computer Science, Multi-Agent Systems

Python

- Active contributor to the development and maintenance of **VITAMIN**, a state-of-the-art model checker
- Designed and implemented model checking procedures for **Intuitionistic Computation Tree Logic (ICTL)** and **Intuitionistic Alternating-time Temporal Logic (IATL)**
- Optimized core functions and improved data structure management, significantly enhancing scalability and performance

## Food Talk

*Sustainability, Software development*

Jun 2023 – Present

*Python, Neo4J*

- Design and development of a system for the traceability of fruit and vegetable product
- The team is planning to collaborate with a leading company to achieve the successful implementation of the project
- The team is planning to collaborate with University of Naples for the advancement of research in this field

## Theater Section

*Human Computer Interaction - Software Development*

Jun 2019 – Feb 2020

*Java, Android, Neo4J*

- Development of a system to respond to the exhibition and conservation needs of “Fondo Niccolini”
- Used Neo4J to represent data because the relationships between them are highly interconnected and complex
- The system is deployed now and operates at “Certosa di San Martino di Napoli”

## EXPERIENCE

---

### Research and Development Internship

*Maipenk s.r.l. | University Federico II, Naples, Italy*

Feb 2023 – Nov 2023

- Conducted research and development efforts for an intraoral scanner.
- Identified the best image processing solutions for the construction of an effective point cloud.

## EDUCATION

---

### Ph.D. in Artificial Intelligence for Agrifood and Environment

*National PhD Program in AI | University Federico II, Naples, Italy*

**Status:** Ongoing

*Nov 2024 – Present*

### M.Sc. in Computer Science, curriculum in Reliable Software Systems

*University Federico II, Naples, Italy*

**Grade:** 110/110 Cum Laude

*27 Sep 2020 – 10 Jul 2024*

### B.Sc. in Computer Science

*University Federico II, Naples, Italy*

**Grade:** 94/110

*25 Sep 2014 – 18 Mar 2020*

## CERTIFICATIONS

---

- Apple Developer Academy
- Agritech Student Academy

## LANGUAGES

---

- Italian - Native Speaker
- English - Intermediate

## AGREEMENTS

---

Autorizzo il trattamento dei dati personali contenuti nel mio curriculum vitae in base all'art. 13 del D. Lgs. 196/2003 e all'art. 13 GDPR 679/16.